Compositional reasoning for Vision

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Why Visual Reasoning Needs Structure

Modern vision-language models (e.g. CLIP, ViLT, Flamingo) are strong *pattern recognizers* but weak *reasoners*.

- More Data
- More Parameters
- More compute

❌ End-to-end models fail or are just compute intensive.

We need compositional, interpretable reasoning beyond end-to-end networks.

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Q) Tag the 7 main characters on the TV show Big Bang Theory

1. Detect Faces
2. Query knowledge
3. Match faces to names
4. Display bounding boxes

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VISPROG Visual Programming for Compositional Visual Reasoning

- A neuro-symbolic system for visual reasoning.
- It uses GPT-3's in-context learning to translate natural language instructions into Python-like programs

Gupta, Tanmay, and Aniruddha Kembhavi. "Visual programming: Compositional visual reasoning without training." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2023.

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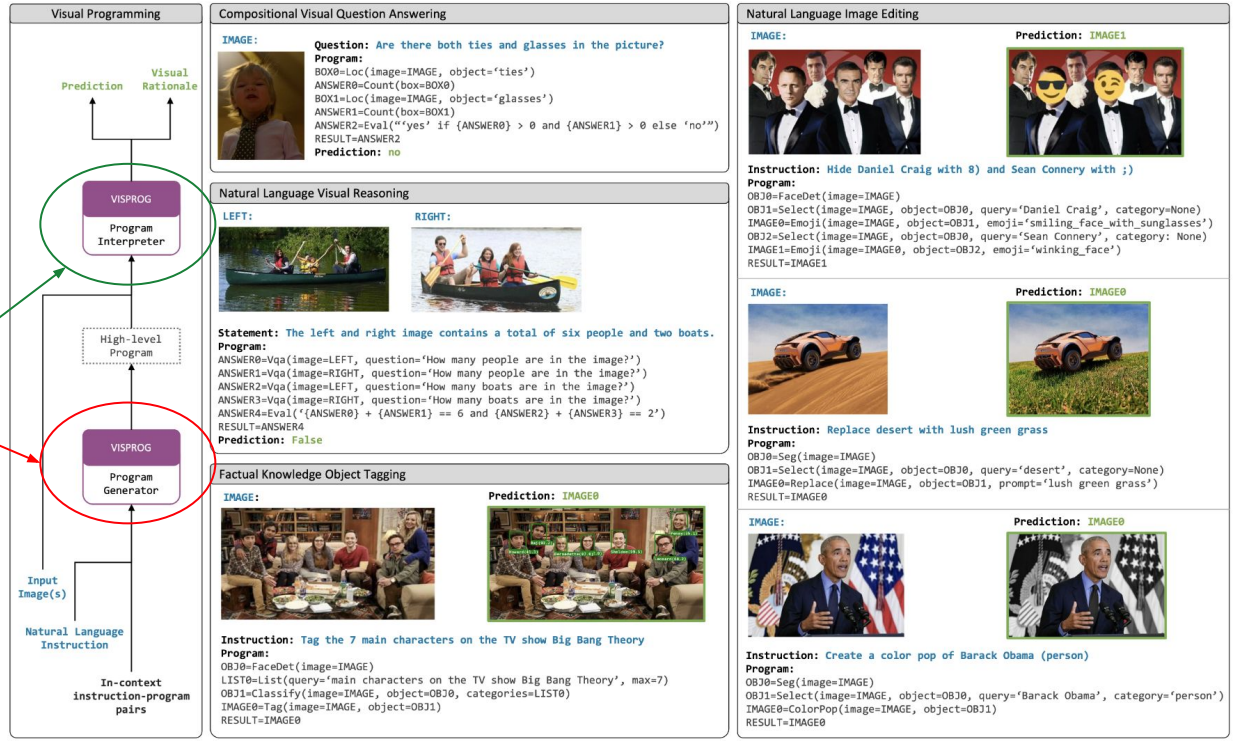
- The program is then executed step-by-step, producing both an answer and a visual rationale.



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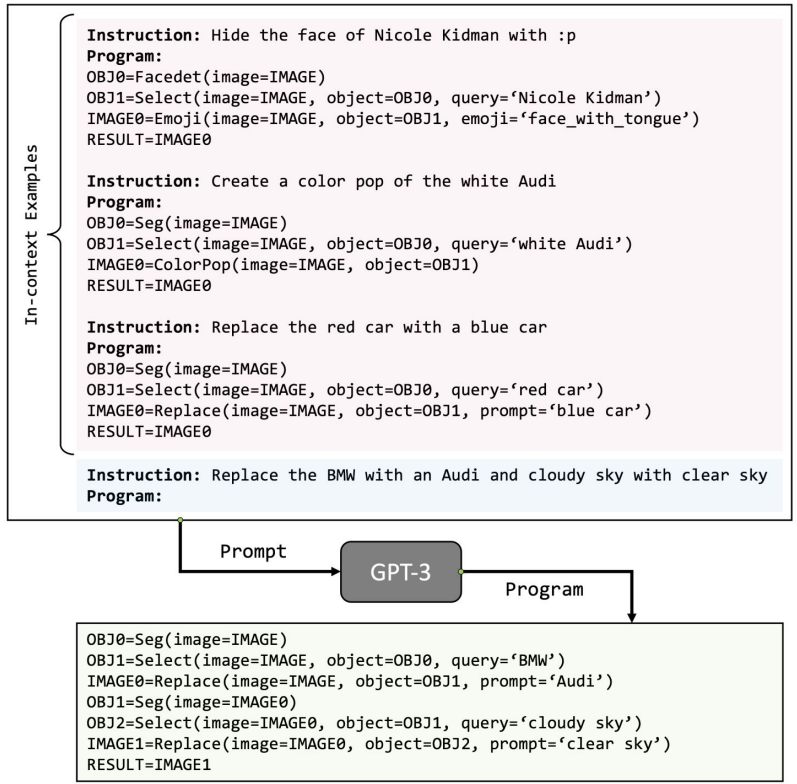
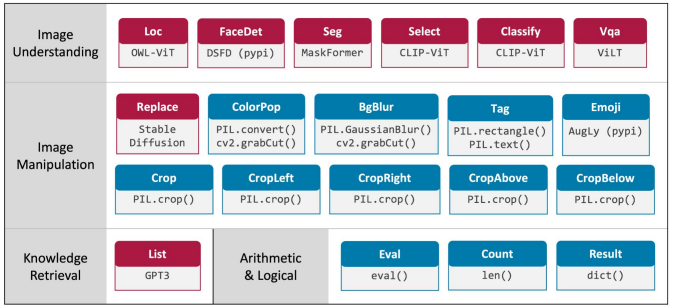


VISPROG Architecture

Input: Instruction + image(s).

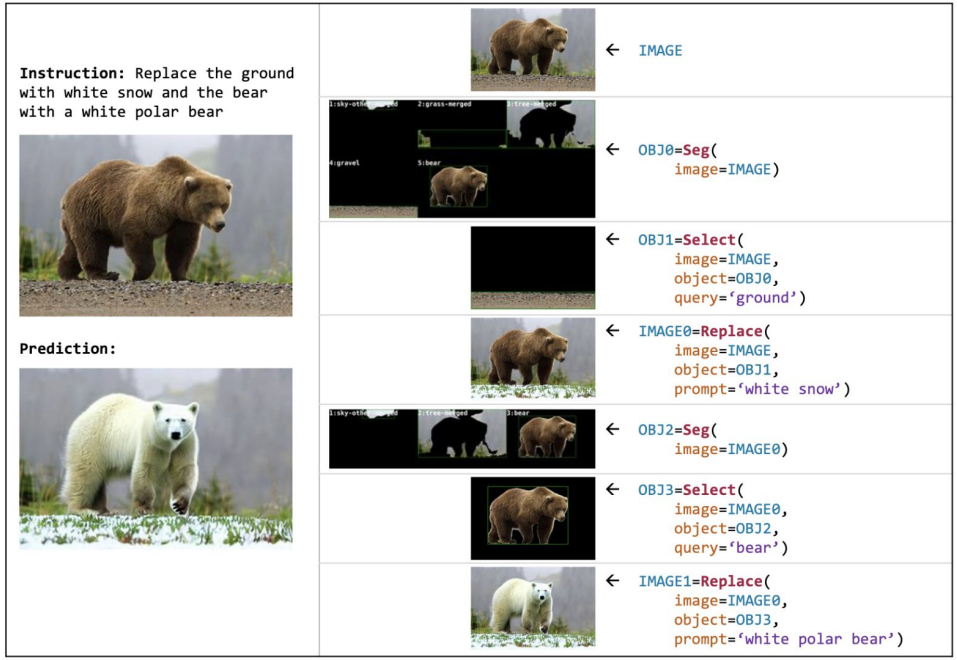
- Step 1: GPT-3 generates a program (sequence of module calls).
- Step 2: VISPROG interpreter executes each module and updates a shared state.
- Step 3: Intermediate outputs are composed into a visual rationale (HTML summary). 20 modules across perception, manipulation, logic.

```
class VisProgModule():
  def __init__(self):
    # load a trained model; move to GPU

  def html(self,inputs: List,output: Any):
    # return an html string visualizing step I/O

  def parse(self,step: str):
    # parse step and return list of input values
    # and variables, and output variable name

  def execute(self,step: str,state: Dict):
    inputs, input_var_names, output_var_name = \
      self.parse(step)

    # get values of input variables from state
    for var_name in input_var_names:
      inputs.append(state[var_name])

    # perform computation using the loaded model
    output = some_computation(inputs)

    # update state
    state[output_var_name] = output

    # visual summary of the step computation
    step_html = self.html(inputs,output)
    return output, step_html
```



⭐ Easy extensibility [add a new module = add a new tool.]

Visual Rationales in Action



Every step's input/output is visible → users can verify reasoning logic.

What can VISPROG do?

- **Compositional VQA (GQA[1])** – + 2.7 pts Acc. vs **ViLT** .
- **Image-Pair Reasoning (NLVRv2)** – 62.4 % zero-shot.
- **Natural Language Image Editing** – uses Stable Diffusion for semantic edits.
- **No fine-tuning** – just in context demonstrations.

Task	Input	Output	Modules
Compositional Visual QA (GQA)	Image + Question	Text	Loc, Vqa, Eval, Count, Crop, CropLeft, CropRight, CropAbove, CropBelow
Reasoning on Image Pairs (NLVR)	Image Pair + Statement	True/False	Vqa, Eval
Factual Knowledge Object Tagging	Image + Instruction	Image	FaceDet, List, Classify, Loc, Tag
Image Editing with Natural Language	Image + Instruction	Image	FaceDet, Seg, Select, Replace, ColorPop, BgBlur, Emoji

[1] https://arxiv.org/abs/1902.09506 [2] https://yoavartzi.com/pub/slya-acl.2017.pdf

What can VISPROG do?

Compositional Visual Question Answering (GQA)	Reasoning on Image Pairs (NLVR²)	Factual Knowledge Object Tagging	Language-Guided Image Editing
• GQA dataset on Accuracy • 50.5 % (voting) +2.7 points over ViLT VQA baseline	• NLVR² dataset Accuracy • 62.4% zero-shot without fine tuning on NLVR²	• 100 instructions/ Custom (46 images) on Precision / Recall / F1 (%) • Tagging F1 = 63.7 → 75.7 (after instruction tuning) • Localization F1 = 80.6% → 84.9% (after instruction tuning)	• 107 instructions/ Custom (65 images) • semantic correctness of the edit measured manually • 59.8 % → 66.4 % (after instruction tuning)

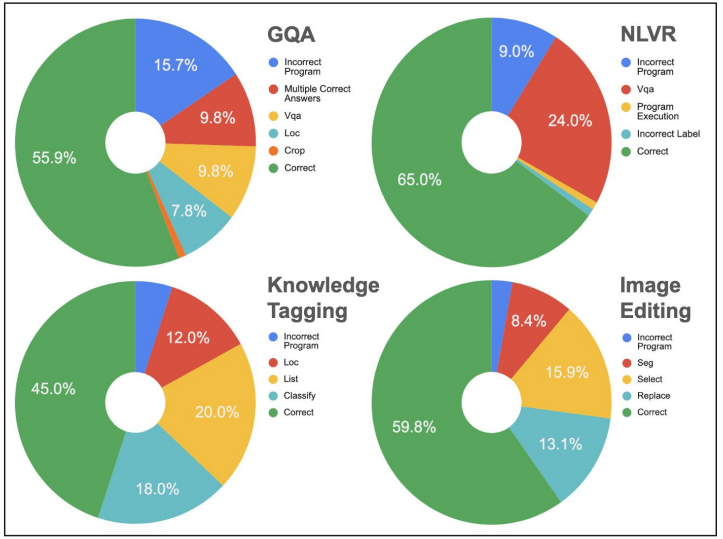
Interpretable, Modular, but Prompt-Dependent

✅ Strengths

- Transparent reasoning (visual rationales).
- Modular and easily extensible (20+ modules).
- Zero-training = adaptable to new tasks quickly.

⚠️ Limitations

- Dependent on prompt examples (manual in-context programming).
- No feedback or self-correction loop.
- Latency: multiple model calls per program.

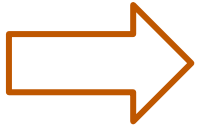


Sources of error analysis on 100 samples for each task

From Visual Programs to Executable Reasoning

VISPROG

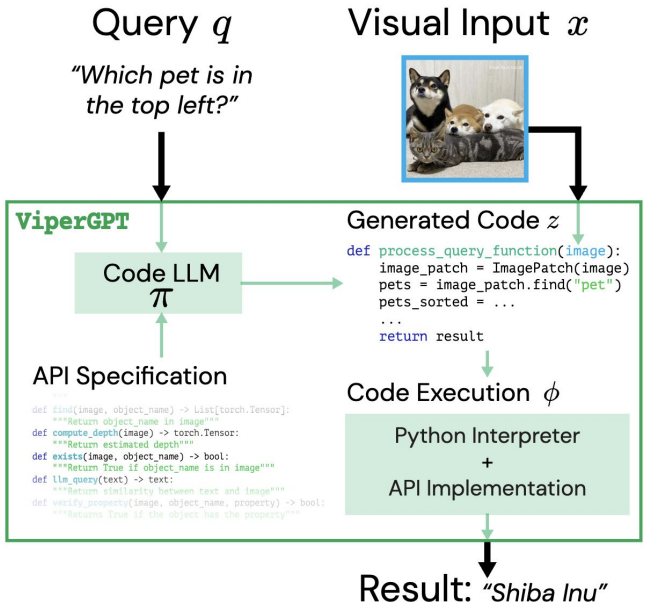
- Custom interpreter (limited control flow)
- Required hand-crafted in-context examples



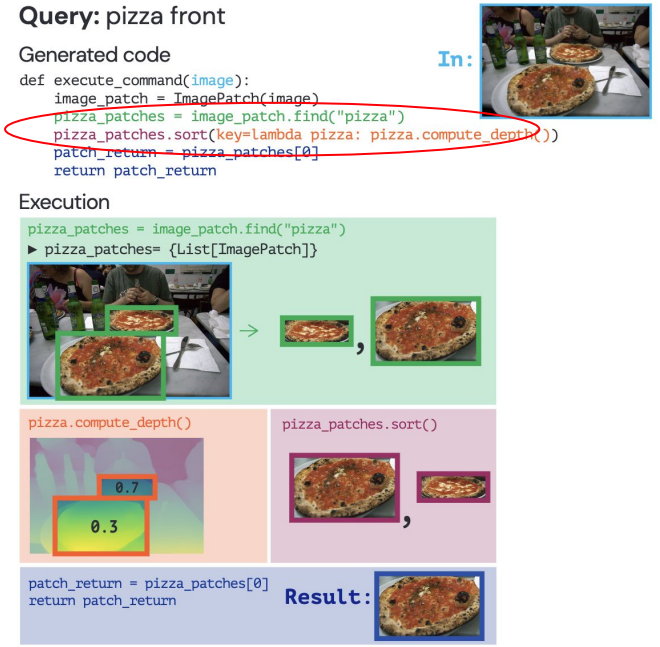
ViperGPT

- Writes real Python code and executes directly using Python interpreter
- Does not use incontext examples but rather function definitions.

Overview



Sample input /output



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API Specification

Specifies:

- Input and output types
- Docstrings
- Examples

❗ Only specifications, no full implementation

- Limited context windows
- Code generation is independent of the changes made to the module implementation

```
def exists(self, object_name: str) -> bool:
    """Returns True if the object specified by object_name is found in the image, and False otherwise.
    Parameters
    -------
    object_name : str
        A string describing the name of the object to be found in the image.

    Examples
    -------
    >>> # Are there both cakes and gummy bears in the photo?
    >>> def execute_command(image)->str:
    >>>     image_patch = ImagePatch(image)
    >>>     is_cake = image_patch.exists("cake")
    >>>     is_gummy_bear = image_patch.exists("gummy bear")
    >>>     return bool_to_yesno(is_cake and is_gummy_bear)
    """
    return len(self.find(object_name)) > 0
```

```
def llm_query(question: str) -> str:
    '''Answers a text question using GPT-3. The input question is always a formatted string with a variable in it.

    Parameters
    ----------
    question: str
        the text question to ask. Must not contain any reference to 'the image' or 'the photo', etc.
    '''
    return llm_query(question)
```

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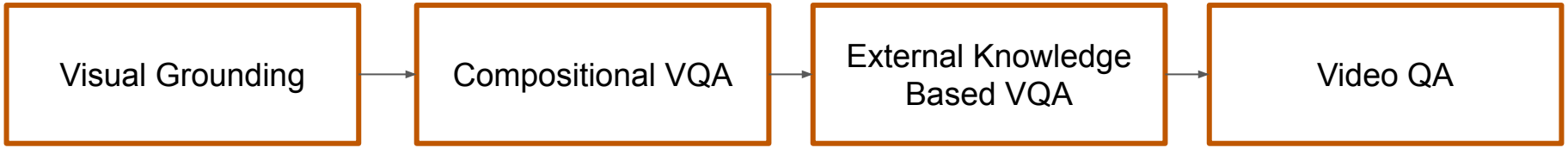
Evaluation on Tasks

- Defines 4 main tasks ranging from basic understanding to complex synthesis
- Each task is a "prerequisite" for following task



- RefCOCO and RefCOCO+ on IoU
- 72.0 / 67.0 +13-17 points ↑ over strongest zero-shot baselines(OWL-ViT, GLIPetc)

- GQA* dataset on Accuracy
- 48.1 % +3.4 points ↑ over BLIP-2.

- OK-VQA dataset on Accuracy
-
- 51.9 % +1.3 points ↑ vs Flamingo; +16 points ↑ vs PNP-VQA.
- First to match SOTA zero-shot knowledge VQA.

- NExT-QA dataset on Accuracy
- 49.8 (T) / 56.4 (C) / 60.0
- Matches or exceeds supervised models.

Query: pizza front

Generated code

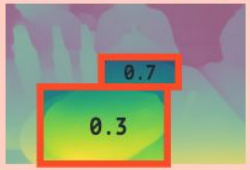
In:

```
def execute_command(image):
    image_patch = ImagePatch(image)
    pizza_patches = image_patch.find("pizza")
    pizza_patches.sort(key=lambda pizza: pizza.compute_depth())
    patch_return = pizza_patches[0]
    return patch_return
```

Execution

```
pizza_patches = image_patch.find("pizza")
▶ pizza_patches= {List[ImagePatch]}
```

pizza.compute_depth()



pizza_patches.sort()

```
patch_return = pizza_patches[0]
return patch_return
```

Result:

sis

xternal Knowledge Based VQA

Video QA

OK-VQA dataset on Accuracy

Better than zero-shot and on-par with some fine-tuned models

- NExT-QA dataset on Accuracy
- Achieves SoTA results, surpassing fine-tuned methods
- frames relations for temporal and causal.

Methods.

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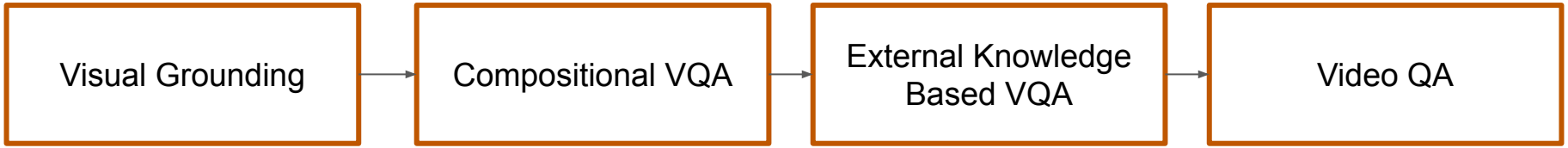

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Evaluation on Tasks

- Defines 4 main tasks ranging from basic understanding to complex synthesis
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Query: Are there water bottles to the right of the bookcase that is made of wood?

In:

Generated code

```
def execute_command(image):
    image_patch = ImagePatch(image)
    bookcase_patches = image_patch.find("bookcase")
    for bookcase_patch in bookcase_patches:
        is_wood = bookcase_patch.verify_property("bookcase", "wood")
        if is_wood:
            water_bottle_patches = image_patch.find("water bottle")
            for water_bottle_patch in water_bottle_patches:
                if water_bottle_patch.horizontal_center > \
                   bookcase_patch.horizontal_center:
                    return "yes"
            return "no"
    return "no"
```

Execution

```
bookcase_patches= image_patch.
      find("bookcase")
▶ bookcase_patches[0] = {ImagePatch}
```

```
▶ bookcase_patches[0].
     horizontal_center = {float} 239.0
...verify_property("bookcase","wood")
▶ is_wood = {bool} True
```

```
water_bottle_patches = image_patch.
      find("water bottle")
▶ water_bottle_patches[0]
   = {ImagePatches}
```

```
▶ water_bottle_patches[0].
     horizontal_center = {float} 608.5
▶ water_bottle_patch.horizontal_center >
bookcase_patch.horizontal_center =
{bool} True
```

Result: "yes"

- Still far behind fine-tuned models
- Decently far behind all fine-tuned Methods.
- models
- frames relations for temporal and causal.

Evaluation on Tasks

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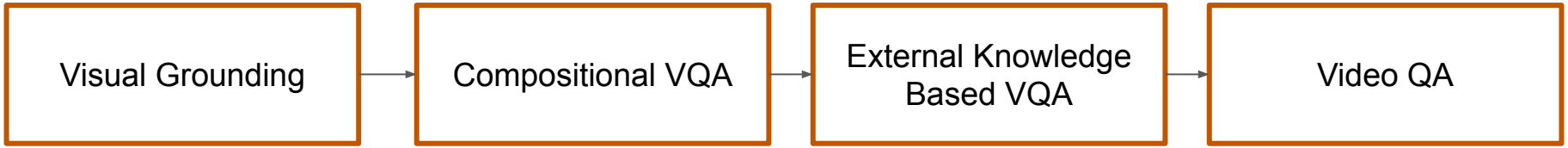

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- Matches or exceeds supervised models.(ATP,HiTeA)

Evaluation on Tasks

Query: What did the boy do after he dropped the sparkles on the floor?

Generated code

```
def execute_command(video, question, possible_answers):
    video_segment = VideoSegment(video)
    drop_detected = False
    for i, frame in enumerate(video_segment.frame_iterator()):
        if frame.exists("boy") and frame.exists("sparkles") and \
                frame.simple_query("is the boy dropping the sparkles?") == "yes":
            drop_detected = True
            break
    if drop_detected:
        index_frame = i + 1
    else:
        index_frame = video_segment.num_frames // 2
    frame_of_interest = ImagePatch(video_segment, index_frame)
    boys = frame_of_interest.find("boy")
    if len(boys) == 0:
        boys = [frame_of_interest]
    boy = boys[0]
    caption = boy.simple_query("What is this?")
    info = {
        "Caption of frame after dropping the sparkles": caption,
    }
    answer = select_answer(info, question, possible_answers)
    return answer
```

Execution

In:

```
frame.exists("boy") and \
frame.exists("sparkles") and \
frame.simple_query("is the boy
   dropping the sparkles?") == "yes":
```

▶ frame = {ImagePatch}

▶ i= {int} 25

```
index_frame = i + 1
```

▶ index_frame = {int} 26

▶ frame_of_interest = {ImagePatch}

```
boys = frame_of_interest.find("boy")
```

▶ boy = {ImagePatch}

▶ caption = {str} "a child running with fire in his hands"

▶ answer = {str} "pick it up"

Result: *"Pick it up"*

fine-tuned models

all fine-tuned Methods.

- frames relations for temporal and causal.

Emergent Capabilities

Query: What would the founder of the brand of the car on the left say to the founder of the brand of the car on the right?


```
def execute_command(image):
    image_patch = ImagePatch(image)
    car_patches = image_patch.find("car")
    car_patches.sort(key=lambda car: car.horizontal_center)
    left_car = car_patches[0]
    right_car = car_patches[-1]
    left_car_brand = left_car.simple_query("What is the brand of this car?")
    right_car_brand = right_car.simple_query("What is the brand of this car?")
    left_car_founder = llm_query(f"Who is the founder of {left_car_brand}?")
    right_car_founder = llm_query(f"Who is the founder of {right_car_brand}?")
    return llm_query(f"What would {left_car_founder} say to {right_car_founder}?")
```

```
car_patches =
image_patch.find("car")
```

```
car_patches.sort(...)
```

```
▶left_car_brand='lamborghini'
▶right_car_brand='ferrari'
```

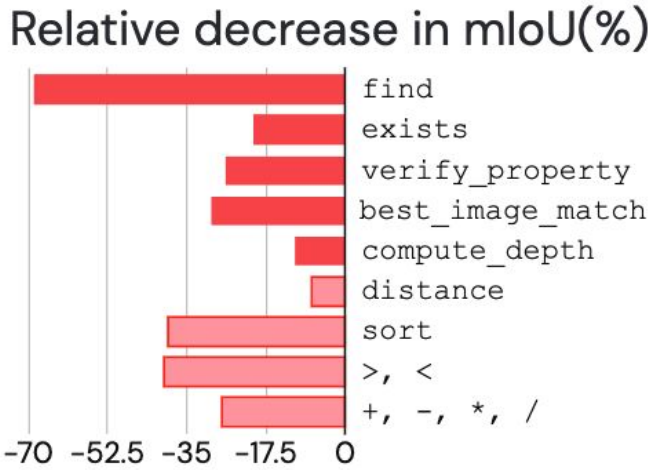
```
▶left_car_founder='Ferruccio Lamborghini'
▶right_car_founder='Enzo Ferrari'
```

Result: *"Ferruccio Lamborghini might say, 'It's been an honor to be a rival of yours for so many years, Enzo. May our cars continue to push each other to be better and faster!' "*

Viper-GPT easily adapts to real-world queries

References: Surís, Dídac, Sachit Menon, and Carl Vondrick. "Vipergpt: Visual inference via python execution for reasoning." ICCV 2023

Intervention | Context-Aware Responses



Query: Return the car that is on the correct lane

```
# Context: the picture was taken in the US
def execute_command(image):
    cars = image.find("car")
    for car in cars:
        if car.horizontal_center > image.horizontal_center:
            return car
    return None
```

Result: *None*

```
# Context: the picture was taken in the UK
def execute_command(image):
    cars = image.find("car")
    for car in cars:
        if car.horizontal_center < image.horizontal_center:
            return car
    return None
```

Result:

References: Surís, Dídac, Sachit Menon, and Carl Vondrick. "Vipergpt: Visual inference via python execution for reasoning." ICCV 2023

Strengths and Weaknesses

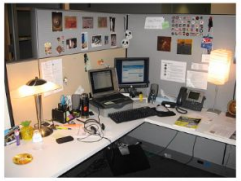
✅ Interpretability Through Explicit Programs
✅ Zero-shot, Task-agnostic Framework
✅ Compositional Reasoning
✅ Modular and Extensible

❌ Heavy Dependence on Code Generation Quality
❌ Reliance on Individual Module Performance
❌ Potentially High Computational and Latency Costs
❌ Lacks general instruction-following capabilities

How do we make these systems reliable?

- VISPROG & ViperGPT = great at modular reasoning, but brittle:
  - ~30% failures = LLM planning errors
  - ~40% = visual module errors (GQA Manual analysis)

Q: Is the keyboard to the left or to the right of the book?



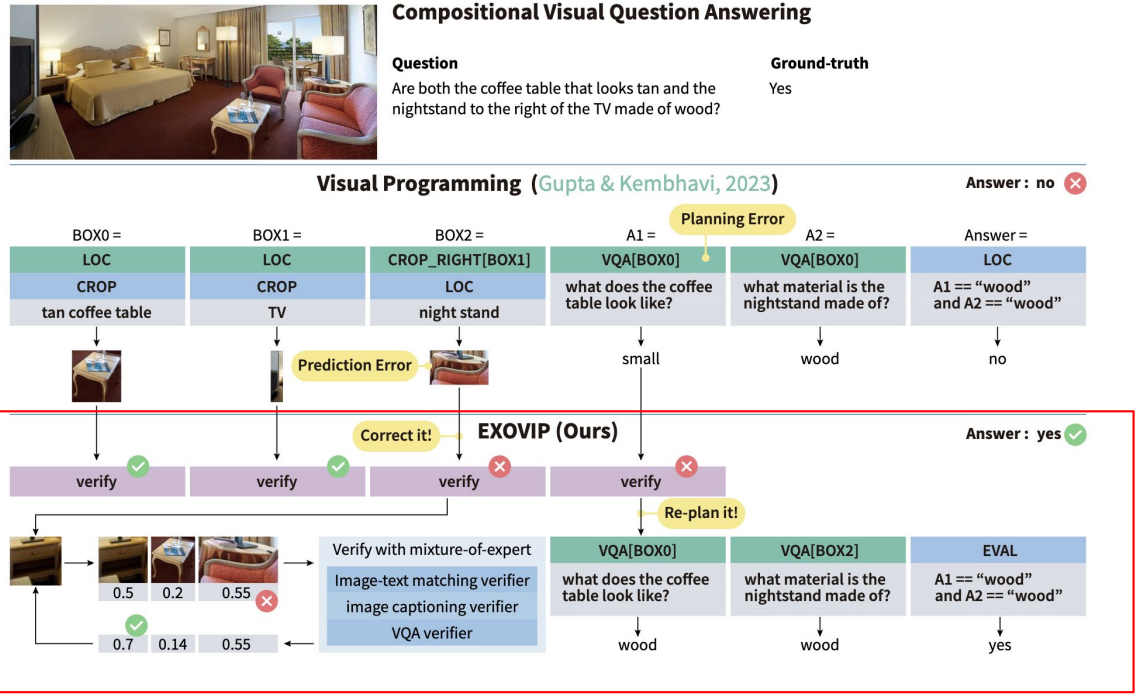
```
BOX0=LOC(image=IMG, object="book")
IMAGE0=CROP(bbox=BOX0)
BOX1=LOC(image= IMAGE0, object="keyboard")
# there is no keyboard in IMAGE0
```

Q: What's the woman holding?

```
BOX0=LOC(image=IMG, object="woman") = NONE
# no box is returned, but there should be one
```

How do we make these systems reliable?

- **ExoViP**[1] proposes a "plug-and-play exoskeleton" that verifies each reasoning step and corrects it.



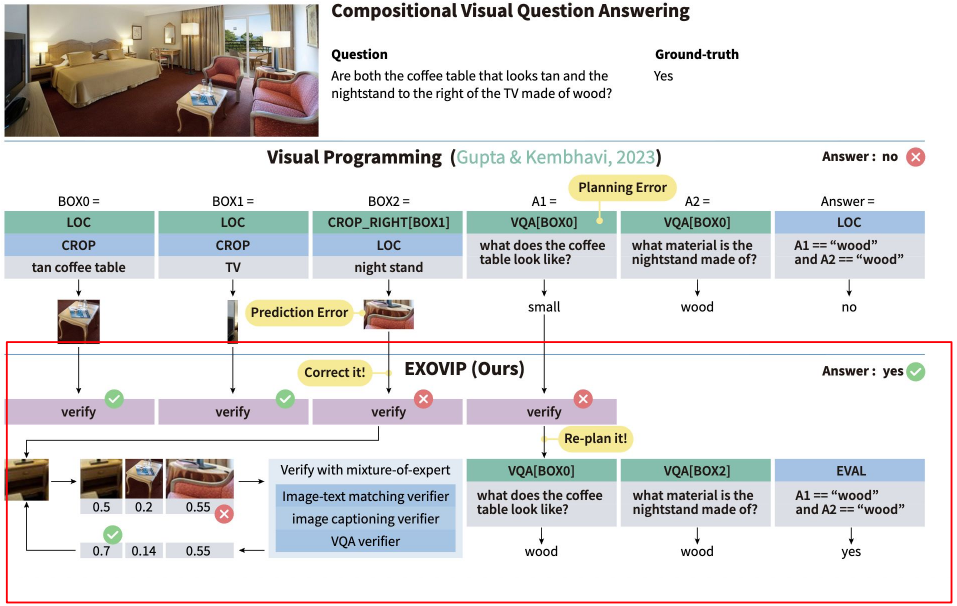
[1] Wang, Yuxuan, et al. "Exovip: Step-by-step verification and exploration with exoskeleton modules for compositional visual reasoning." arXiv preprint arXiv:2408.02210 (2024).

Verification as an Exoskeleton

ExoViP adds **verification modules** that “wrap around” each reasoning step.
These *exoskeletons* check correctness *after every step*, using three “sub-verifiers”:

- **Image-Text Matching Verifier (CLIP-based)** — checks if result visually fits description.
- **Image Captioning Verifier (BLIP)** — compares captioned scene to expectation.
- **VQA Verifier** — asks yes/no validation questions about each output.

Each produces a *verification score* → system adjusts or re-plans if inconsistency found.



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Verification as an Exoskeleton

Initialize Tree:
Start with LLM-generated program; each child = next module step.

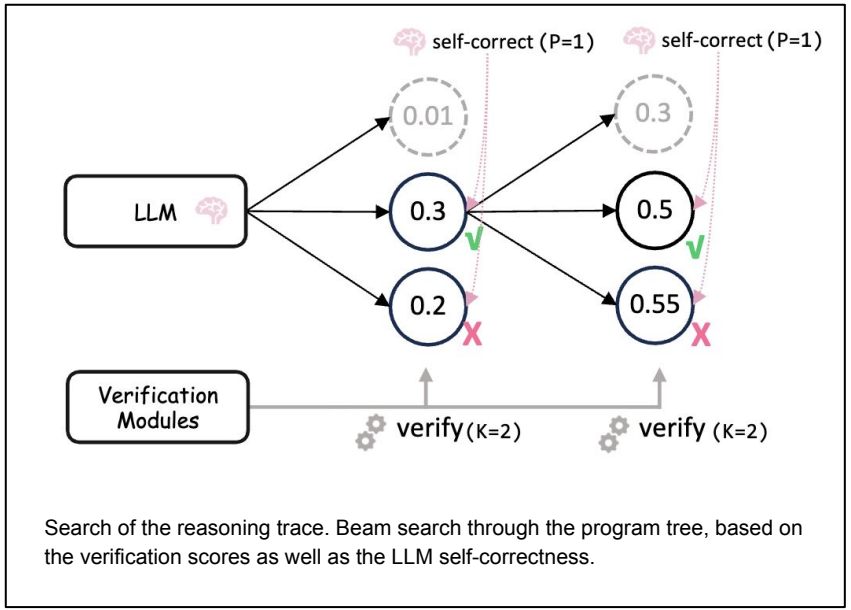
Execute & Verify:
Run each step → get verifier scores → normalize as weights.

Calibrate:
Combine module confidence & verifier weight → $p' = w \times p$.

Beam Search:
Keep top-K traces each expansion (branching ≈ 3).

Post-hoc Self-Correction (PSC):
If scores similar → LLM ranks traces → keep top-P (P < K).

Select Final Trace:
Choose path with highest combined verification + self-consistency score.



Search of the reasoning trace. Beam search through the program tree, based on the verification scores as well as the LLM self-correctness.

[1] Wang, Yuxuan, et al. "Exovip: Step-by-step verification and exploration with exoskeleton modules for compositional visual reasoning." arXiv preprint arXiv:2408.02210 (2024).

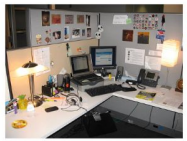
Mechanism: “Verify, Calibrate, Re-plan”

Q: What's the woman holding?

```
BOX0=LOC(image=IMG, object="woman") = NONE
# no box is returned, but there should be one
# after verification, this error is corrected
IMAGE0=CROP(bbox=BOX0)
ANSWER0=VQA(image=IMAGE0,
            question="what is the woman holding?")
```

(a) module error

Q: Is the keyboard to the left or to the right of the book?



```
BOX0=LOC(image=IMG, object="book")
IMAGE0=CROP(bbox=BOX0)
BOX1=LOC(image= IMAGE0, object="keyboard")
# there is no keyboard in IMAGE0
# after verification, the program is corrected:
# CROP ⟶ CROP_RIGHT
ANSWER0=COUNT(bbox= BOX1)
ANSWER1=EVAL(expression="'left'
             if ANSWER0 > 0  else 'right'")
```

(b) planning error

[1] Wang, Yuxuan, et al. "Exovip: Step-by-step verification and exploration with exoskeleton modules for compositional visual reasoning." arXiv preprint arXiv:2408.02210 (2024).

Results

Applied on **VISPROG** and **ViperGPT** pipelines → consistent gains:

- On **GQA** compositional VQA: from 57.4 → 61.5 accuracy (+4%).
- On **RefCOCO**: from 27.3 → 31.5 IoU (+4.2).
- Also improved abstract reasoning, image editing, and video reasoning tasks.
- Works as a universal add-on that improves both VISPROG and ViperGPT without retraining.

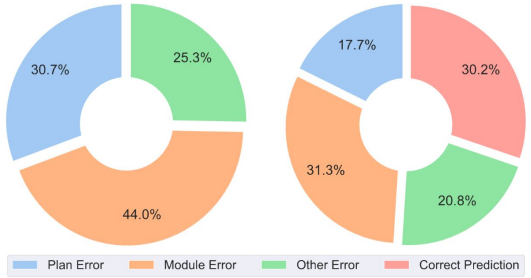
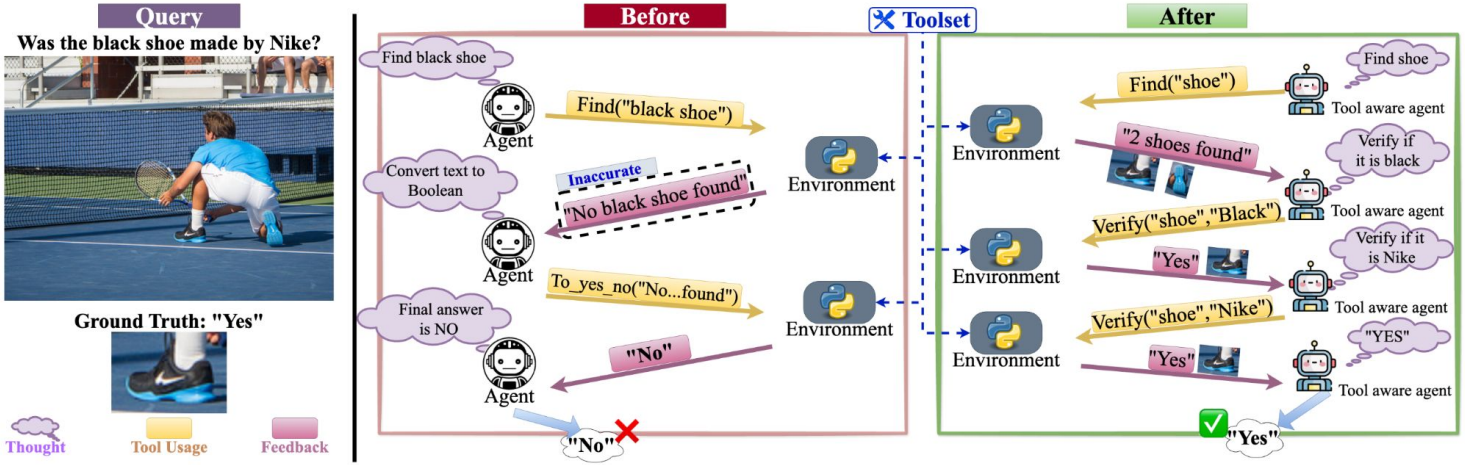


Figure 4: Distribution of the failure cases of original VISPROG (left), and distribution of the failure cases of EXOVIP (right)

DWIM (ICCV 2025): Teaching Agents to Use Tools Wisely

- Vis Prog, ViperGpt and Even self-correcting agents (ExoViP) don’t truly understand tool behavior.
- DWIM Train LLMs to become tool-aware, recognizing when tools fail and improving their use.

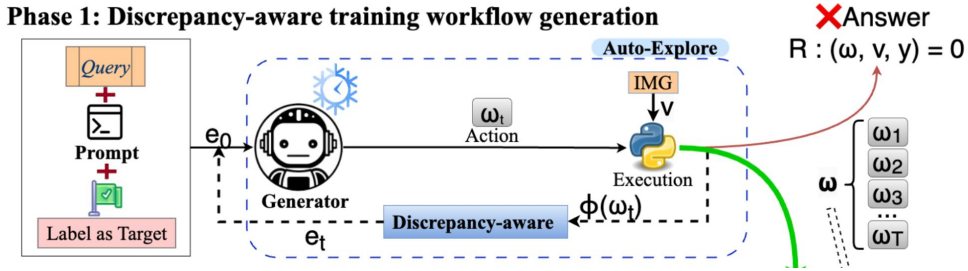


Discrepancy-Aware Workflow Generation

Standard compositional agents assume every tool output is correct.
DWIM adds “discrepancy detection”:

- Compares expected answer vs tool feedback.
- If mismatch → generates a “Rethink” action (natural-language note explaining tool failure + next plan).
- Collects these as training workflows that include both successes and failures.

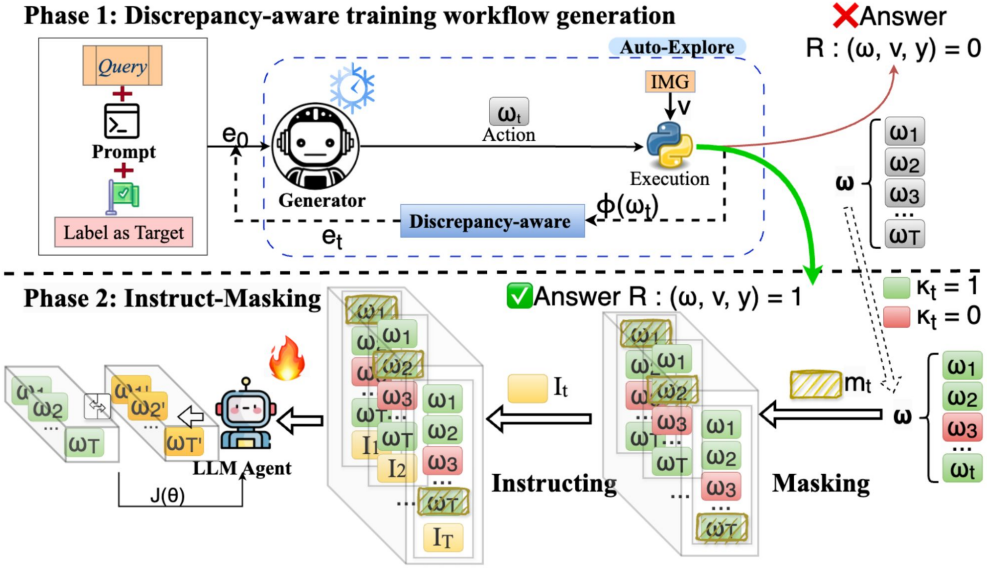
This creates richer supervision, not just right answers, but why wrong steps failed.



Instruct-Masking Fine-Tuning

DWIM trains the LLM to copy only effective actions.

- Each workflow step is labeled as effective or ineffective.
- During training, DWIM masks only the effective actions and instructs the model to regenerate them.
- This prevents the model from cloning noisy or incorrect tool usage.
- Essentially:
  - Naive SFT: copies everything (good + bad).
  - DWIM: copies only what worked — becoming tool-aware.
- Inspired by BERT's masking + instruction-tuning.



Results & Impact

Performance

- +9.7 % avg. improvement over HYDRA/VisRep.
- GQA 69.3 %, OK-VQA 62.8 %, SugarCREPE 74.6 %.

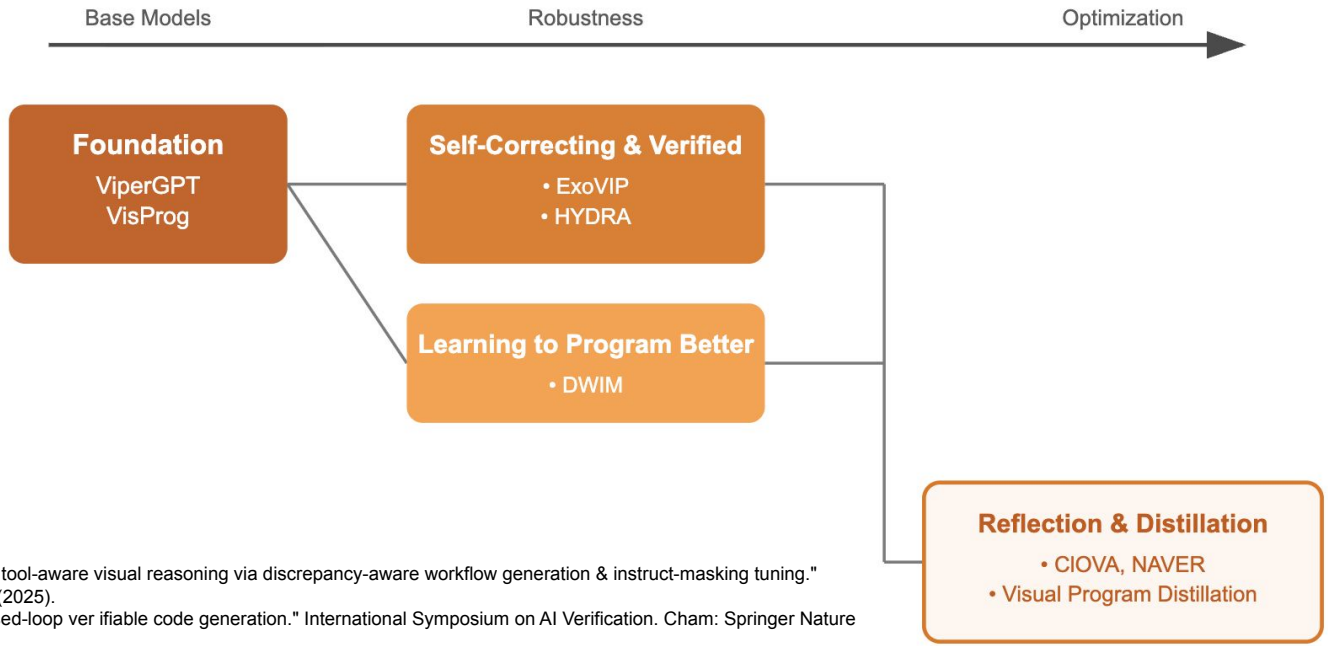
Generalization :

- Cross-dataset transfer → +4–7 % gain.

Tool efficiency:

- Average tool calls ↓ from ~3.3 (HYDRA) → 1.7 (DWIM).

From Programs to Agents and Learners



Ke, Fucai, et al. "Dwim: Towards tool-aware visual reasoning via discrepancy-aware workflow generation & instruct-masking tuning." arXiv preprint arXiv:2503.19263 (2025).
Sun, Chuyue, et al. "Clover: Closed-loop ver ifiable code generation." International Symposium on AI Verification. Cham: Springer Nature Switzerland, 2024.
Cai, Zhixi, et al. "Naver: A neuro-symbolic compositional automaton for visual grounding with explicit logic reasoning." arXiv preprint arXiv:2502.00372 (2025).
Hu, Yushi, et al. "Visual program distillation: Distilling tools and programmatic reasoning into vision-language models." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024.

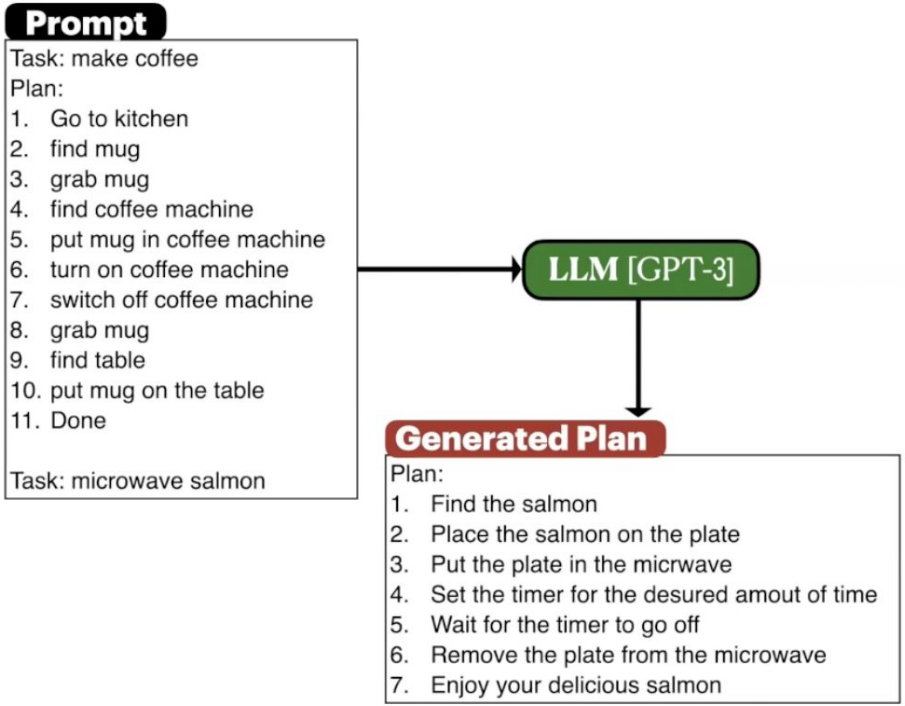
Questions?

PROGPROMPT: Generating Situated Robot Task Plans using Large Language Models

Ishika Singh, Valts Blukis, Arsalan Mousavian, Ankit Goyal, Danfei Xu, Jonathan Tremblay, Dieter Fox, Jesse Thomason, Animesh Garg

ICRA 2023

Background - LLMs for Task Planning

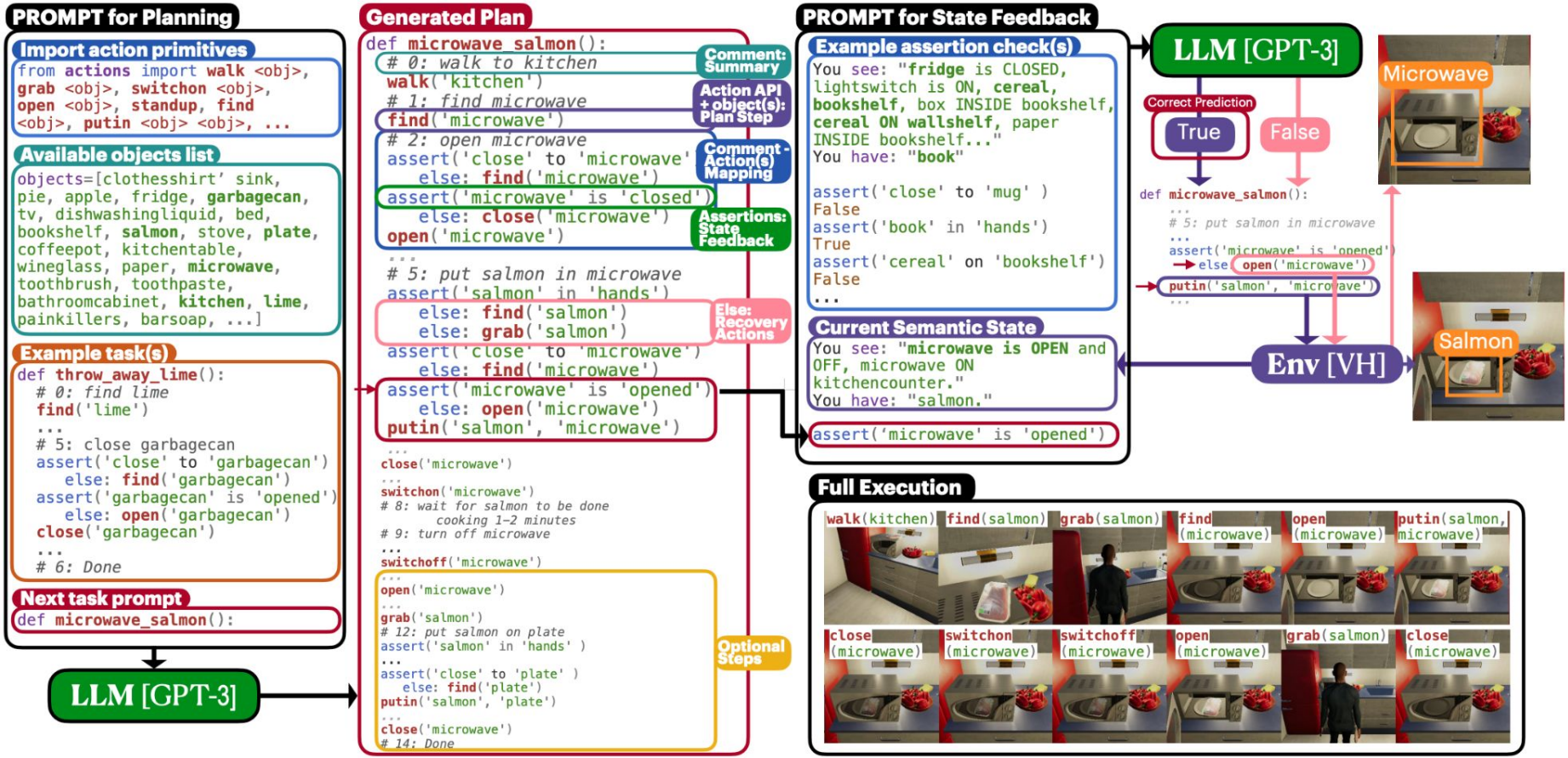


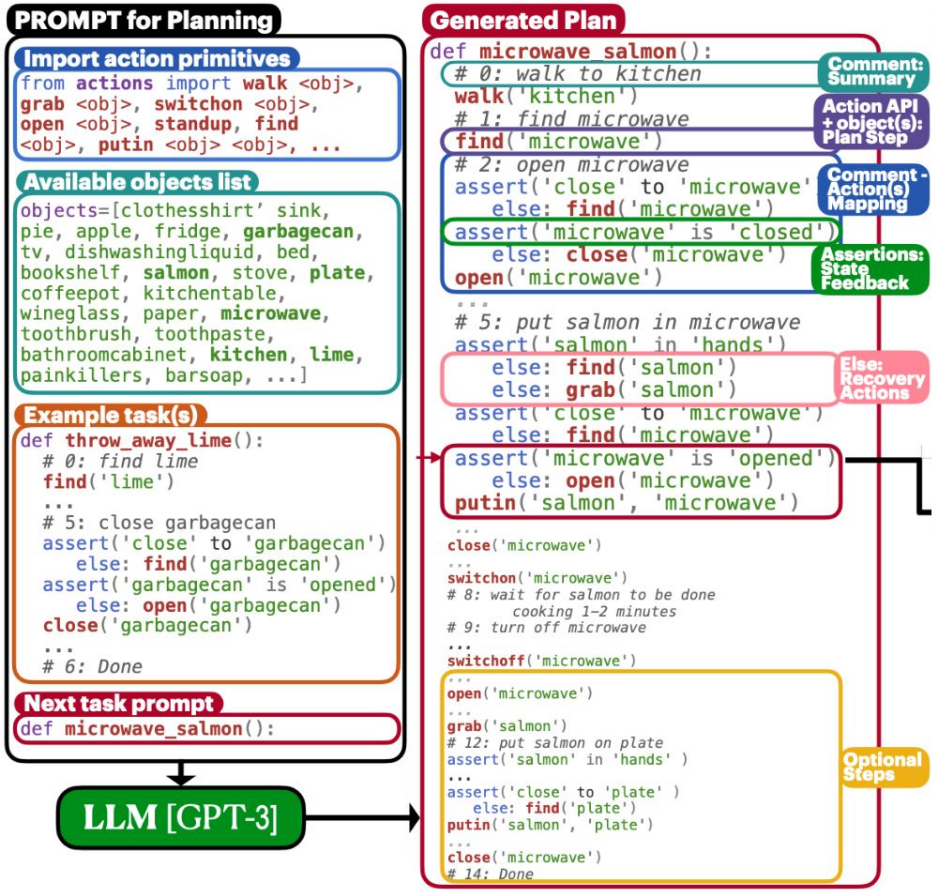
Challenges:

- LLM is not situated in the scene
- Text-to-robot action mapping may not be trivial
- Plan Steps using unavailable actions and objects
- Combinatorial admissible action space

Solution

Instead of using natural language, prompt the LLM with program-like specifications of the available actions and objects in an environment


PROMPT for Planning
Import action primitives
from actions import walk <obj>, grab <obj>, switchon <obj>, open <obj>, standup, find <obj>, putin <obj> <obj>, ...
Available objects list
objects=[clothesshirt' sink, pie, apple, fridge, garbagecan, tv, dishwashingliquid, bed, bookshelf, salmon, stove, plate, coffeepot, kitchentable, wineglass, paper, microwave, toothbrush, toothpaste, bathroomcabinet, kitchen, lime, painkillers, barsoap, ...]
Example task(s)
def throw_away_lime():
# 0: find lime
find('lime')
...
# 5: close garbagecan
assert('close' to 'garbagecan')
else: find('garbagecan')
assert('garbagecan' is 'opened')
else: open('garbagecan')
close('garbagecan')
...
6: Done
Next task prompt
def microwave_salmon():
LLM [GPT-3]
Generated Plan
def microwave_salmon():
# 0: walk to kitchen
walk('kitchen')
# 1: find microwave
find('microwave')
# 2: open microwave
assert('close' to 'microwave')
else: find('microwave')
assert('microwave' is 'closed')
else: close('microwave')
open('microwave')
...
# 5: put salmon in microwave
assert('salmon' in 'hands')
else: find('salmon')
else: grab('salmon')
assert('close' to 'microwave')
else: find('microwave')
assert('microwave' is 'opened')
else: open('microwave')
putin('salmon', 'microwave')
...
close('microwave')
...
switchon('microwave')
# 8: wait for salmon to be done
cooking 1-2 minutes
9: turn off microwave
...
switchoff('microwave')
...
open('microwave')
...
grab('salmon')
# 12: put salmon on plate
assert('salmon' in 'hands')
...
assert('close' to 'plate')
else: find('plate')
putin('salmon', 'plate')
...
close('microwave')
# 14: Done
Comment: Summary
Action API + object(s): Plan Step
Comment - Action(s) Mapping
Assertions: State Feedback
Else: Recovery Actions
Optional Steps
PROMPT for State Feedback
Example assertion check(s)
You see: "fridge is CLOSED, lightswitch is ON, cereal, bookshelf, box INSIDE bookshelf, cereal ON wallshelf, paper INSIDE bookshelf..."
You have: "book"
assert('close' to 'mug')
False
assert('book' in 'hands')
True
assert('cereal' on 'bookshelf')
False
...
Current Semantic State
You see: "microwave is OPEN and OFF, microwave ON kitchencounter."
You have: "salmon."
assert('microwave' is 'opened')
LLM [GPT-3]
Correct Prediction
True
False
def microwave_salmon():
...
5: put salmon in microwave
...
assert('microwave' is 'opened')
else open('microwave')
putin('salmon', 'microwave')
...
Env [VH]
Microwave
Salmon
Full Execution
walk(kitchen)
find(salmon)
grab(salmon)
find (microwave)
open (microwave)
putin(salmon, microwave)
close (microwave)
switchon (microwave)
switchoff (microwave)
open (microwave)
grab(salmon)
close (microwave)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# 2: open microwave
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else: find('microwave')
assert('microwave' is 'closed')
else: close('microwave')
open('microwave')
...
# 5: put salmon in microwave
assert('salmon' in 'hands')
else: find('salmon')
else: grab('salmon')
assert('close' to 'microwave')
else: find('microwave')
assert('microwave' is 'opened')
else: open('microwave')
putin('salmon', 'microwave')
...
close('microwave')
...
switchon('microwave')
# 8: wait for salmon to be done
cooking 1-2 minutes
9: turn off microwave
...
switchoff('microwave')
...
open('microwave')
...
grab('salmon')
# 12: put salmon on plate
assert('salmon' in 'hands')
...
assert('close' to 'plate')
else: find('plate')
putin('salmon', 'plate')
...
close('microwave')
# 14: Done
Comment: Summary
Action API + object(s): Plan Step
Comment - Action(s) Mapping
Assertions: State Feedback
Else: Recovery Actions
Optional Steps

PROMPT for Planning

Import action primitives

```
from actions import walk <obj>,
grab <obj>, switchon <obj>,
open <obj>, standup, find
<obj>, putin <obj> <obj>, ...
```

Available objects list

```
objects=[clothesshirt' sink,
pie, apple, fridge, garbagecan,
tv, dishwashingliquid, bed,
bookshelf, salmon, stove, plate,
coffeepot, kitchentable,
wineglass, paper, microwave,
toothbrush, toothpaste,
bathroomcabinet, kitchen, lime,
painkillers, barsoap, ...]
```

Example task(s)

```
def throw_away_lime():
  # 0: find lime
  find('lime')
  ...
  # 5: close garbagecan
  assert('close' to 'garbagecan')
     else: find('garbagecan')
  assert('garbagecan' is 'opened')
     else: open('garbagecan')
  close('garbagecan')
  ...
  # 6: Done
```

Next task prompt

```
def microwave_salmon():
```

LLM [GPT-3]

Generated Plan

```
def microwave_salmon():
 # 0: walk to kitchen
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    else: close('microwave')
 open('microwave')
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    else: find('microwave')
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    else: open('microwave')
 putin('salmon', 'microwave')
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 close('microwave')
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 # 8: wait for salmon to be done
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 # 9: turn off microwave
 ...
 switchoff('microwave')
 ...
 open('microwave')
 ...
 grab('salmon')
 # 12: put salmon on plate
 assert('salmon' in 'hands' )
 ...
 assert('close' to 'plate' )
    else: find('plate')
 putin('salmon', 'plate')
 ...
 close('microwave')
 # 14: Done
```

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Action API + object(s): Plan Step

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Assertions: State Feedback

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Optio Steps

PROMPT for State Feedback

Example assertion check(s)

```
You see: "fridge is CLOSED,
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bookshelf, box INSIDE bookshelf,
cereal ON wallshelf, paper
INSIDE bookshelf..."
You have: "book"

assert('close' to 'mug' )
False
assert('book' in 'hands')
True
assert('cereal' on 'bookshelf')
False
...
```

Current Semantic State

```
You see: "microwave is OPEN and
OFF, microwave ON
kitchencounter."
You have: "salmon."
```

```
assert('microwave' is 'opened')
```

LLM [GPT-3]

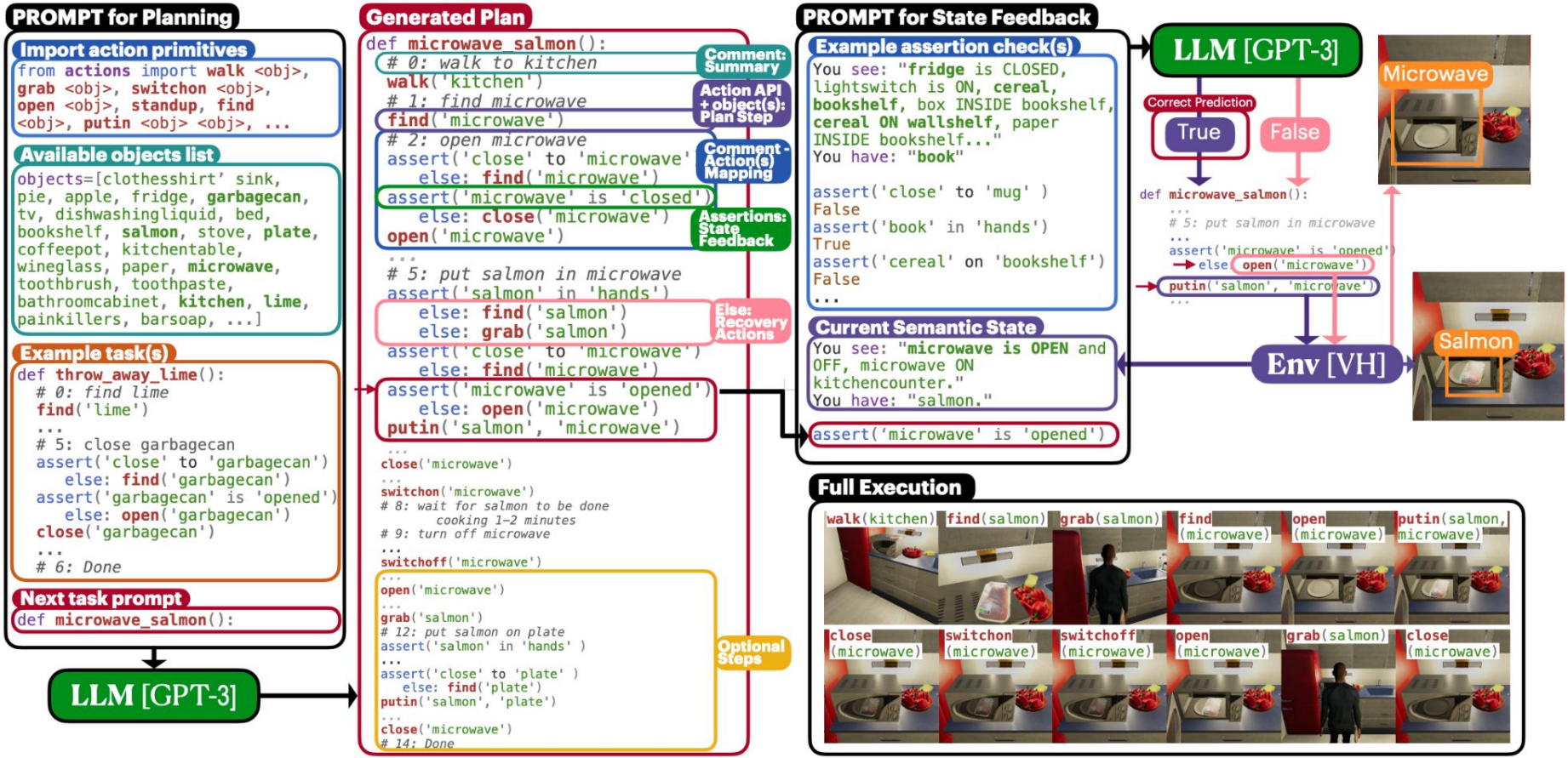
Correct Prediction

True

False

```
def microwave_salmon():
   ...
   # 5: put salmon in microwave
   ...
   assert('microwave' is 'opened')
      else open('microwave')
   putin('salmon', 'microwave')
   ...
```

Env [VH]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else: grab('salmon')
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else: find('microwave')
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else: open('microwave')
putin('salmon', 'microwave')
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You have: "book"
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False
assert('book' in 'hands')
True
assert('cereal' on 'bookshelf')
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Env [VH]
Microwave
Salmon
Full Execution
walk(kitchen)
find(salmon)
grab(salmon)
find (microwave)
open (microwave)
putin(salmon, microwave)
close (microwave)
switchon (microwave)
switchoff (microwave)
open (microwave)
grab(salmon)
close (microwave)

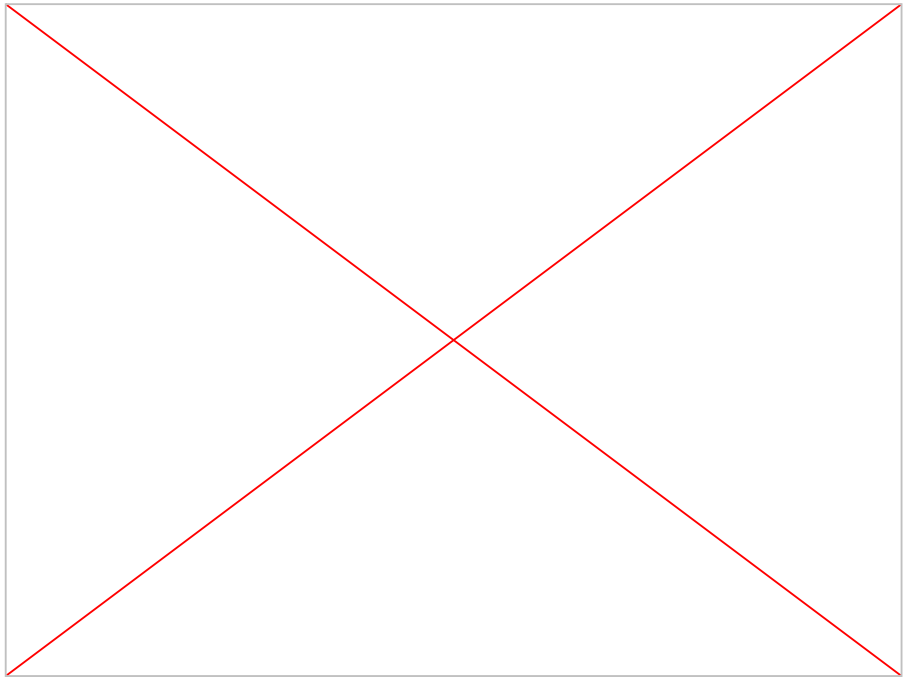
Experiment - Evaluation Metrics

- Success Rate (SR): the fraction of executions that achieved all task-relevant goal-conditions
- Goal Conditions Recall (GCR): the percentage of critical goals that were actually accomplished
- Executability (Exec): the fraction of actions in the generated plan that are actually executable by the robot in the environment

Experiment - Simulation Experiments

Virtual Home (VH) Environment

- A VH state s is a set of objects O and properties P.
- P encodes information like in(salmon, microwave) and agent close to(salmon).
- VH provides observations in the form of state graph with object properties and relations



Example task: put salmon in the fridge

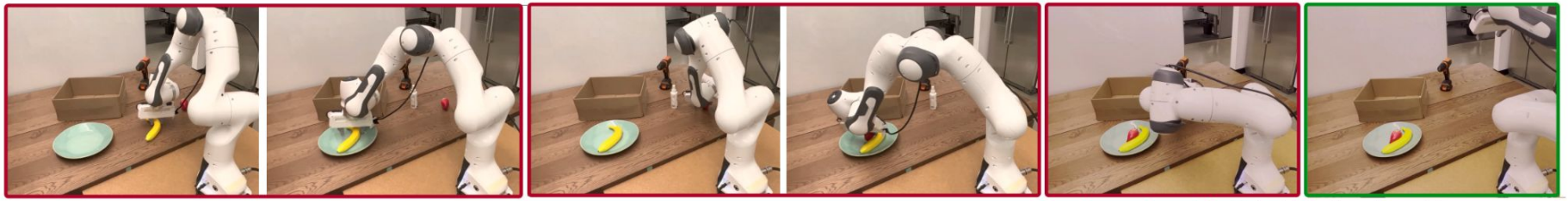
#	— Prompt Format and Parameters — Format	COMMENTS	FEEDBACK	LLM Backbone	*SR*	*Exec*	*GCR*
1	PROGPROMPT	✓	✓	CODEX	**0.40**±0.11	**0.90**±0.05	**0.72**±0.09
2	PROGPROMPT	✓	✓	DAVINCI	0.22±0.04	0.60±0.04	0.46±0.04
3	PROGPROMPT	✓	✓	GPT3	0.34±0.08	0.84±0.01	0.65±0.05
4	PROGPROMPT	✓	✗	GPT3	0.28±0.04	0.82±0.01	0.56±0.02
5	PROGPROMPT	✗	✓	GPT3	0.30±0.00	0.65±0.01	0.58±0.02
6	PROGPROMPT	✗	✗	GPT3	0.18±0.04	0.68±0.01	0.42±0.02
7	LANGPROMPT	-	-	GPT3	0.00±0.00	0.36±0.00	0.42±0.02
8	Baseline from HUANG ET AL. [2]			GPT3	0.00±0.00	0.45±0.03	0.21±0.03

Observation:

- PROGPROMPT outperforms prior work
- CODEX exceeds GPT3 performance
- FEEDBACK improve performance
- Removing COMMENTS from the prompt code substantially reduces performance

Experiment - Real-Robot Experiments

Task: sort fruits on the plate and bottles in the box



`grab_and_puton('banana', 'plate')` `grab_and_puton('strawberry', 'plate')` `grab_and_putin('bottle', 'box')`

- No feedback mechanism: Physical robot setup did not allow reliably tracking system state and checking assertions, and is prone to random failures due to things like grasps slipping.
- Introduce an additional metric:
- Plan SR: whether the plan would have likely succeeded, provided successful pick-and-place execution without gripper failures.

Task Description	Distractors	*SR*	Plan *SR*	*GCR*
put the banana in the bowl	0	1	1	1/1
	4	1	1	1/1
put the pear on the plate	0	1	1	1/1
	4	1	1	1/1
put the banana on the plate and the pear in the bowl	0	1	1	2/2
	3	1	1	2/2
sort the fruits on the plate and the bottles in the box	0	0	1	2/3
	1	1	1	3/3
	2	0	0	2/3

Observation:

- Across tasks, with and without distractor objects, the system almost always succeeds, failing only on the sort task.

Strengths and Weaknesses

✅ Long-Horizon Planning
✅ Generalization to New Tasks and Environments
✅ Executable and Valid Task Plans
✅ Error Recovery via Precondition Checking (Feedback)

❌ Environment interaction during execution is limited to the commonsense precondition checking-based feedback
❌ Focus on Semantic State Only
❌ Limited Information Exchange (Text-Only Modality)

Questions?